Resources & Further Reading

Immersive Reader

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A curated list of papers, tools, data sources, and references behind the Agentic Forecasting bootcamp. Organized by content type; each entry has a one-line note on how it relates to the project. Papers link to their **arXiv** versions.

> **How to read this list.** The bootcamp asks two questions: *can LLMs and agents forecast time series effectively?* and *how do advances in agentic AI apply to forecasting?* The papers below supply the evidence and the techniques; the tools and data sources are what the reference implementations are built from.

Research papers

Can LLMs and agents forecast? — benchmarks & evidence

- **ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities** — Karger, Tetlock et al. (2024). arXiv:2409.19839 · live leaderboard A continuously-updated benchmark of *future* (unresolved) questions, so there's no data leakage. Headline finding: expert humans still beat the best LLMs. The honest baseline for "how good are LLM forecasters, really?"
- **Wisdom of the Silicon Crowd: LLM Ensemble Prediction Rivals Human Crowd Accuracy** — Schoenegger et al. (2024). arXiv:2402.19379 · Science Advances An ensemble of 12 LLMs matches the human crowd on probabilistic questions — motivation for treating LLMs as aggregatable probabilistic forecasters.

LLM Processes & context-grounded forecasting

- **LLM Processes: Numerical Predictive Distributions Conditioned on Natural Language** — Requeima, Bronskill, Choi, Turner, Duvenaud (NeurIPS 2024). arXiv:2405.12856 · code The seminal LLM-Process paper: elicit coherent probabilistic forecasts from an LLM, conditioned on numbers *and* natural-language priors. The conceptual basis for our `llm_processes` predictor family.

- **Context is Key (CiK): A Benchmark for Forecasting with Essential Textual Information** — Williams et al. (ICML 2025). arXiv:2410.18959 · PMLR Tasks where text is *required* to forecast well. Directly motivates report-grounded prompting (CFPR documents, BoC releases) — and §5.4 is the covariate-serialization pattern our `SampledTrajectoryLLMPredictor` uses.

Self-improving & self-adaptive agents — the arc behind our Adaptive Agent

This trio (all from Jeff Clune's group, incl. Vector affiliation) forms one arc: agents that search over their own programs, memory, or strategy. It's the research context for our energy **Adaptive Agent**.

- **Automated Design of Agentic Systems (ADAS)** — Hu, Lu, Clune (ICLR 2025). arXiv:2408.08435 · code A "meta agent" writes new agents *in code*, iterating on past discoveries. The idea that agent design can itself be automated.
- **Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents** — Zhang, Hu, Lu, Lange, Clune (2025). arXiv:2505.22954 · code A self-improving system that edits its own code and keeps changes that *empirically* help, exploring via an archive of past solutions.
- **Learning to Continually Learn via Meta-learning Agentic Memory Designs (ALMA)** — Xiong, Hu, Clune (ICLR 2026 Workshop RSI). arXiv:2602.07755 · project Meta-learns the agent's *memory design* (schema + retrieval/update) as code — the generalization of "let the agent learn how to learn" that our adaptive agent gestures at with its mutable strategy state.

Foundational forecasting

- **Forecasting: Principles and Practice (3rd ed.)** — Hyndman & Athanasopoulos. otexts.com/fpp3 The standard, free, online forecasting textbook. The reference for the classical methods (ETS, ARIMA, etc.) we benchmark against.
- **Strictly Proper Scoring Rules, Prediction, and Estimation** — Gneiting & Raftery (2007). doi:10.1198/016214506000001437 The theory behind CRPS / Brier / RPS — why we score *distributions*, not point estimates, throughout the evaluation harness.

Core tools & libraries (used in the repo)

- **Darts** — docs · GitHub The primary numerical-forecasting library; backs our `numerical` predictors (AutoARIMA, ETS, Kalman, LightGBM, linear regression).
- **Prophet** — docs Trend/seasonality decomposition baseline; the statistical floor in the energy/WTI capability staircase.
- **Google Agent Development Kit (ADK)** — adk.dev · GitHub The agent framework behind our agentic forecasters (`build_adk_agent`, runners, skills, code-execution tools).
- **Langfuse** — docs LLM/agent tracing and observability; our backbone for inspecting traces and for the reasoning-alignment (LLM-as-judge) evaluation.
- **E2B** — docs Cloud sandbox for agent code execution — the `run_code` tool agents use to compute indicators and self-calibrate intervals.
- **LiteLLM** — docs Unified LLM API layer; how predictors route to models through the Vector proxy.

Data sources

- **Statistics Canada** — statcan.gc.ca · CPI table 18-10-0004 Canadian CPI sub-indices (getting-started gasoline, food CPI) and the policy-rate series (BoC). Carries publication lag — central to our cutoff discipline.
- **FRED (Federal Reserve Economic Data)** — fred.stlouisfed.org · API Macro/commodity covariates (rates, spreads, unemployment, oil/gold). Used by the S&P 500 and BoC implementations. Free personal API key required.
- **Yahoo Finance** — finance.yahoo.com · via `https://github.com/ranaroussi/yfinance` Equities, indices, and futures (`^GSPC`, `^VIX`, `CL=F`). Source for the S&P 500 and WTI series.
- **Bank of Canada** — rate decisions · press releases · Monetary Policy Report Rate-decision calendar + published rationales — the target and the cutoff-aware context/judge material for the BoC discrete-event implementation.
- **Canada's Food Price Report (CFPR)** — Dalhousie Agri-Food Analytics Lab. 2026 edition · lab home The annual food-inflation forecast our food-CPI implementation replicates; the report PDFs are extracted as cutoff-aware LLM-Process context.

For inspiration — *not built into the repo*

These are pointers for build-phase projects and discussion, not dependencies of the reference implementations.

Agent & skill optimizers

- **DSPy: Compiling Declarative LM Calls into Self-Improving Pipelines** — Khattab et al. (ICLR 2024). arXiv:2310.03714 · dspy.ai · GitHub Treat prompts/pipelines as programs an optimizer compiles. An automated alternative to hand-tuning the prompt strategies our predictors use by hand.
- **SkillOpt: Executive Strategy for Self-Evolving Agent Skills** — Microsoft (2026). arXiv:2605.23904 · project · GitHub Optimizes a skill markdown file as the trainable "parameter" of a frozen agent — closely parallel to our Adaptive Agent's learned, mutable strategy skill.

Time-series foundation models

Zero-shot pretrained forecasters — a natural build-phase extension (add one as a new `Predictor`).

- **TimesFM** (Google) — GitHub
- **Chronos** (Amazon) — GitHub
- **Moirai / uni2ts** (Salesforce) — GitHub